

Sum and difference of cubes



1 -5

2

a $(x + y)(x^2 - xy + y^2)$

b $(x - y)(x^2 + xy + y^2)$

c $(x + 1)(x^2 - x + 1)$

d $(x - 2)(x^2 + 2x + 4)$

e $(x + 3)(x^2 - 3x + 9)$

f $(x - 4)(x^2 + 4x + 16)$

g $(x + 8)(x^2 - 8x + 64)$

h $(x - 10)(x^2 + 10x + 100)$

3

a $(x + 4y)(x^2 - 4xy + 16y^2)$

b $(x - 5y)(x^2 + 5xy + 25y^2)$

c $(x + 9y)(x^2 - 9xy + 81y^2)$

d $(x - 6y)(x^2 + 6xy + 36y^2)$

e $(x + 10y)(x^2 - 10xy + 100y^2)$

f $(a + 3b)(a^2 - 3ab + 9b^2)$

g $(s + 5t)(s^2 + 5st + 25t^2)$

h $(u - 7v)(u^2 + 7uv + 49v^2)$

4

a $(3x + 4)(9x^2 - 12x + 16)$

b $(3x - 2)(9x^2 + 6x + 4)$

c $(10x + 9)(100x^2 - 90x + 81)$

d $(8x - 7)(64x^2 + 56x + 49)$

e $(2x + 10)(4x^2 - 20x + 100)$

f $(5x - 1)(25x^2 + 5x + 1)$

g $(8x + 2)(64x^2 - 16x + 4)$

h $(6x - 4)(36x^2 + 24x + 16)$

5

a $2(5u + 3v)(25u^2 - 15uv + 9v^2)$

b $2(3m - 5n)(9m^2 + 15mn + 25n^2)$

c $(3x + 2y)(9x^2 - 6xy + 4y^2)$

d $(5x - 4y)(25x^2 + 20xy + 16y^2)$

e $(6x + 7y)(36x^2 - 42xy + 49y^2)$

f $(7x - 10y)(49x^2 + 70xy + 100y^2)$

g $5(2u + v)(4u^2 - 2uv + v^2)$

h $4(m - 2n)(m^2 + 2mn + 4n^2)$

6

a $(2x - 1)(x^2 - x + 169)$

b $-5(3x^2 - 3x + 7)$

c $(2x + 3)(x^2 + 3x + 39)$

d $8(3x^2 - 18x + 43)$

7

a $25x^2 - 10x + 4$

b $\frac{x^2 + 5x + 25}{4}$

c $\frac{x^2 + 5x + 25}{x + 5}$

d $4x^2 + 6xy + 9y^2$

8



a $(x - y)(x^2 + xy + y^2)(x + y)(x^2 - xy + y^2)$

b $(x - y)(x + y)(x^4 + x^2y^2 + y^4)$

c $(x^2 + xy + y^2)(x^2 - xy + y^2)$

9 1547